

Marino_Presentation

Recurring Win Profiles in a Pooled Five-Map Valorant Dataset
CSCI 111 Machine Learning Project: Classification and Clustering

CSCI 111 Final Presentation

Central Question

In a pooled dataset of Ascent, Haven, Bind, Icebox, and Split, what recurring win-related profiles appear, and how are they associated with composition flags?

Important framing: this is map-level outcome modeling from same-map aggregate data, not pure pre-match forecasting.

Why This Question Matters

Valorant Context

- Five-versus-five tactical shooter played round by round on one map.
- Attackers take space and plant the spike; defenders stop or retake.
- Pro teams win through mechanics, agent composition, utility, map control, and execution.

Measured Signals

- **Composition structure:** roles brought into a map.
- **Carry shape:** how performance is distributed across a team.
- The question is association, not proof that one strategy always wins.

Role translation: Duelists create entry pressure; Initiators gather information; Controllers provide smokes and map control; Sentinels anchor sites or flanks.

Project focus: whether composition and carry-shape signals are associated with recurring win-related profiles in professional Valorant maps.

Rubric Alignment

CSCI 111 Requirement	V4 Implementation
Labeled dataset	Binary label: Team_A_Win.
Dataset size	Final cleaned dataset has 22,793 rows, well above 500.
Numeric inputs	19 official numeric inputs.
D1 / D2 / D3	Original, MinMax-scaled, and Standard-scaled versions.
Two classification methods	Logistic Regression and Random Forest.
Six classification models	Each classifier run on D1, D2, and D3.
Clustering	K-Means on label-dropped D1, D2, and D3.
Recommendation and conclusions	Best model selected and final interpretation stated with limitations.

Rubric-Safe Claim Boundary

Results are reported as associations in this pooled dataset. The deck does not claim causal meta conclusions or pure pre-match prediction.

Dataset Construction

Source Pipeline

- Raw local exports: `vct_2021` through `vct_2026`.
- `maps_scores.csv`: map outcomes.
- `overview.csv`: player-map stats and selected agents.
- Public source citation: Ryan Luong, *Valorant Champion Tour 2021-2026 Data*, Kaggle.

Final rows

22,793

Retained maps

Ascent, Haven, Bind, Icebox, Split

Label

Team_A_Win

Cleaning Logic

Keep full-map player rows, require exactly five players per team-map, require both teams to join successfully, and retain only the five core maps.

Official Feature Set and D1 / D2 / D3

19 Official Inputs

- Year
- Map dummies: Ascent, Bind, Haven, Icebox; Split is reference.
- Team A and Team B carry shape: Team_StdDev, Star_Gap.
- Team A and Team B composition flags: Double Duelist, Double Initiator, Double Controller, Double Sentinel, No Sentinel.

```
D1 = official_df[feature_columns + [label_column]].copy()
```

```
minmax = MinMaxScaler()  
D2[feature_columns] = minmax.fit_transform(D1[feature_columns])
```

```
standard = StandardScaler()  
D3[feature_columns] = standard.fit_transform(D1[feature_columns])
```

Standard scaling centers features around 0 with comparable spread. D2 and D3 preserve the same 19 inputs and label.

Classification Setup

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

models = {
    "Logistic Regression": LogisticRegression(
        max_iter=3000, random_state=42),
    "Random Forest": RandomForestClassifier(
        n_estimators=100, random_state=42)
}
```

Evaluation Design

- Classification = supervised prediction of Team_A_Win.
- Same stratified split reused across the six official models.
- For supervised evaluation, D2 and D3 scaling is fit on the training fold only.
- Metrics: accuracy, precision, recall, and confusion matrices.

Models

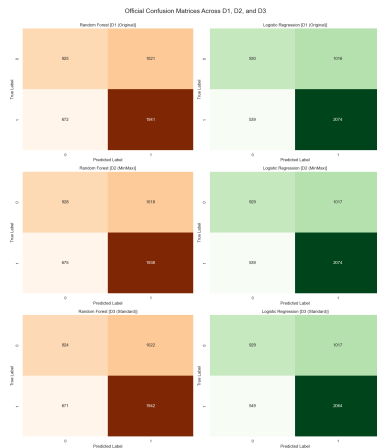
2 classifiers × 3 datasets = 6

Classification Results and Recommendation

Model	Accuracy	Precision	Recall
D1 Random Forest	0.6286	0.6553	0.7428
D1 Logistic Regression	0.6589	0.6712	0.7937
D2 Random Forest	0.6286	0.6556	0.7417
D2 Logistic Regression	0.6587	0.6710	0.7937
D3 Random Forest	0.6286	0.6552	0.7432
D3 Logistic Regression	0.6565	0.6699	0.7899

Recommendation

Majority baseline is **0.5732**. D1 Logistic Regression has the highest raw accuracy, but D3 Logistic Regression is recommended because it is effectively tied within 0.005 and better matches normalized linear-model practice.



Saved

six-model confusion-matrix figure from the V4 notebook.

Map-by-Map Composition Tendencies

Map	Strongest Common Flag	Win Rate	Weakest Common Flag	Win Rate
Ascent	Double Sentinel	0.5452	Double Duelist	0.4863
Haven	Double Sentinel	0.5179	No Sentinel	0.4747
Bind	Double Controller	0.5274	Double Sentinel	0.4728
Icebox	Double Sentinel	0.5161	Double Initiator	0.4644
Split	Double Controller	0.5112	Double Initiator	0.4701

Interpretation

This is a scouting-style descriptive table, not causal proof. Some role structures look more favorable on specific maps, but no single composition flag is universally best across all five maps.

Clustering Setup

```
X_cluster = dataset.drop(columns=[label_column])
```

```
for k in range(2, 8):  
    km = KMeans(  
        n_clusters=k,  
        random_state=42,  
        n_init=10  
    )  
    labels = km.fit_predict(X_cluster)  
    score = silhouette_score(sample_X,  
                             sample_labels)
```

Rubric-Safe Clustering

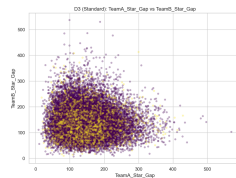
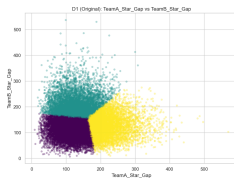
- K-Means is run on D1, D2, and D3 after dropping Team_A_Win.
- Clustering = grouping maps without using the win label.
- Candidate clusters: $k = 2..7$.
- Best k is selected by silhouette score.
- Win rates are joined back only after clustering for post-hoc description.

Clustering Results

Dataset	Best k	Silhouette
D1 Original	3	0.3501
D2 MinMax	7	0.2334
D3 Standard	2	0.2112

Read

D1 gives the clearest view by silhouette; D2 and D3 show the pattern is scaling-sensitive.



Scatter: Team A star gap vs Team B star gap. Labels are heuristic, not official esports archetypes.

Final Answer and Limitations

Direct Answer

In this pooled five-map dataset, recurring composition and carry-shape profiles appear. Their win associations are descriptive, scaling-sensitive, and not causal proof of one universal meta.

Main Findings

- Best official recommendation: **D3 Logistic Regression**.
- D3 LR metrics: 0.6565 accuracy, 0.6699 precision, 0.7899 recall.
- Higher-win and lower-win clusters can both include carry-shaped profiles.

Limits to State Clearly

- 2021 and 2022 contribute about 91.6% of retained rows.
- Map-level same-map aggregates, not pure pre-match forecasting.
- Cite Kaggle VCT 2021-2026 and submit local V4 files for reproducibility.